

```
1 package com.astromedicomp.win_rtr3;
2
3 import android.content.Context; // for drawing context related
4 import android.opengl.GLSurfaceView; // for OpenGL Surface View and all related
5 import javax.microedition.khronos.opengles.GL10; // for OpenGL ES 1.0 needed as param type GL10
6 import javax.microedition.khronos.egl.EGLConfig; // for EGLConfig needed as param type EGLConfig
7 import android.opengl.GLES32; // for OpenGL ES 3.2
8 import android.view.MotionEvent; // for "MotionEvent"
9 import android.view.GestureDetector; // for GestureDetector
10 import android.view.GestureDetector.OnGestureListener; // OnGestureListener
11 import android.view.GestureDetector.OnDoubleTapListener; // for OnDoubleTapListener
12
13 // A view for OpenGL ES3 graphics which also receives touch events
14 public class GLESView extends GLSurfaceView implements GLSurfaceView.Renderer, OnGestureListener, OnDoubleTapListener
15 {
16     private final Context context;
17
18     private GestureDetector gestureDetector;
19
20     public GLESView(Context drawingContext)
21     {
22         super(drawingContext);
23
24         context=drawingContext;
25
26         // accordingly set EGLContext to current supported version of OpenGL-ES
27         setEGLContextClientVersion(3);
28
29         // set Renderer for drawing on the GLSurfaceView
30         setRenderer(this);
31
32         // Render the view only when there is a change in the drawing data
33         setRenderMode(GLSurfaceView.RENDERMODE_WHEN_DIRTY);
34
35         gestureDetector = new GestureDetector(context, this, null, false); // this means 'handler' i.e. who is going to handle
36         gestureDetector.setOnDoubleTapListener(this); // this means 'handler' i.e. who is going to handle
37     }
38
39     // overridden method of GLSurfaceView.Renderer ( Init Code )
40     @Override
41     public void onSurfaceCreated(GL10 gl, EGLConfig config)
42     {
43         // OpenGL-ES version check
44         String version = gl.glGetString(GL10.GL_VERSION);
45         System.out.println("VDG: "+version);
46
47         initialize(gl);
48     }
49 }
```

```
49
50 // overridden method of GLSurfaceView.Renderer ( Chnge Size Code )
51 @Override
52 public void onSurfaceChanged(GL10 unused, int width, int height)
53 {
54     resize(width, height);
55 }
56
57 // overridden method of GLSurfaceView.Renderer ( Rendering Code )
58 @Override
59 public void onDrawFrame(GL10 unused)
60 {
61     draw();
62 }
63
64 // Handling 'onTouchEvent' Is The Most IMPORTANT,
65 // Because It Triggers All Gesture And Tap Events
66 @Override
67 public boolean onTouchEvent(MotionEvent e)
68 {
69     // code
70     int eventaction = e.getAction();
71     if(!gestureDetector.onTouchEvent(e))
72         super.onTouchEvent(e);
73     return(true);
74 }
75
76 // abstract method from OnDoubleTapListener so must be implemented
77 @Override
78 public boolean onDoubleTap(MotionEvent e)
79 {
80     System.out.println("VDG: "+"Double Tap");
81     return(true);
82 }
83
84 // abstract method from OnDoubleTapListener so must be implemented
85 @Override
86 public boolean onDoubleTapEvent(MotionEvent e)
87 {
88     // Do Not Write Any code Here Because Already Written 'onDoubleTap'
89     return(true);
90 }
91
92 // abstract method from OnDoubleTapListener so must be implemented
93 @Override
94 public boolean onSingleTapConfirmed(MotionEvent e)
95 {
96     System.out.println("VDG: "+"Single Tap");
97     return(true);
98 }
99
100 // abstract method from OnGestureListener so must be implemented
101 @Override
102 public boolean onDown(MotionEvent e)
103 {
104     // Do Not Write Any code Here Because Already Written
```

```
'onSingleTapConfirmed'
105     return(true);
106 }
107
108 // abstract method from OnGestureListener so must be implemented
109 @Override
110 public boolean onFling(MotionEvent e1, MotionEvent e2, float velocityX,  ↗
    float velocityY)
111 {
112     return(true);
113 }
114
115 // abstract method from OnGestureListener so must be implemented
116 @Override
117 public void onLongPress(MotionEvent e)
118 {
119     System.out.println("VDG: "+"Long Press");
120 }
121
122 // abstract method from OnGestureListener so must be implemented
123 @Override
124 public boolean onScroll(MotionEvent e1, MotionEvent e2, float distanceX,  ↗
    float distanceY)
125 {
126     System.out.println("VDG: "+"Scroll");
127     System.exit(0);
128     return(true);
129 }
130
131 // abstract method from OnGestureListener so must be implemented
132 @Override
133 public void onShowPress(MotionEvent e)
134 {
135 }
136
137 // abstract method from OnGestureListener so must be implemented
138 @Override
139 public boolean onSingleTapUp(MotionEvent e)
140 {
141     return(true);
142 }
143
144 private void initialize(GL10 gl)
145 {
146     // Set the background frame color
147     GLES32.glClearColor(0.0f, 0.0f, 1.0f, 1.0f);
148 }
149
150 private void resize(int width, int height)
151 {
152     // Adjust the viewport based on geometry changes,
153     // such as screen rotation
154     GLES32.glViewport(0, 0, width, height);
155 }
156 public void draw()
157 {
```

```
158         // Draw background color
159         GLES32.glClear(GLES32.GL_COLOR_BUFFER_BIT |
                        GLES32.GL_DEPTH_BUFFER_BIT);
160     }
161 }
162
```

